

A Proof of Bala's Strong Divisibility Conjecture for $F(k^n) - 1$, OEIS A000071

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Abstract

Let $a(j) = F(j) - 1$, sequence A000071 of the OEIS. In December 2022, P. Bala conjectured that for every odd positive integer k the sequence $(a(k^n))_{n \geq 1}$ is a strong divisibility sequence, that is $\gcd(a(k^n), a(k^m)) = a(k^{\gcd(n,m)})$ for all $n, m \geq 1$. We prove it. The proof rests on a single uniform factorisation: for odd j one has $F(j) - 1 = F(E)L(O)$, where E and O are respectively the even and the odd member of the pair $\{(j-1)/2, (j+1)/2\}$. Applying this to $j = k^n$ turns the conjecture into a bookkeeping problem about $\gcd(k^n \pm 1, k^m \pm 1)$ and the classical gcd laws for Fibonacci and Lucas numbers; four cases, distinguished by $k \bmod 4$ and by the parities of n and m , then close it. The hypothesis that k be odd is essential and we show exactly where it enters.

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1 The conjecture

Let $F(0) = 0$, $F(1) = 1$, $F(j+2) = F(j+1) + F(j)$ and let L denote the companion Lucas sequence, $L(0) = 2$, $L(1) = 1$. Sequence A000071 of the OEIS [2] is

$$a(j) = F(j) - 1 = 0, 0, 0, 1, 2, 4, 7, 12, 20, 33, 54, \dots$$

On 5 December 2022 Peter Bala contributed the comment:

Conjecture: for k a positive odd integer, the sequence $\{a(k^n) : n \geq 1\}$ is a strong divisibility sequence; that is, for $n, m \geq 1$, $\gcd(a(k^n), a(k^m)) = a(k^{\gcd(n,m)})$.

The comment stands unrebutted on the live entry; the page carried the timestamp 22 August 2026. Note that the strong divisibility is asserted in the *exponent*: the index of the sequence $b(n) := a(k^n)$ is n , and $\gcd(b(n), b(m)) = b(\gcd(n, m))$. It is not the statement that a itself is a strong divisibility sequence, which is false.

Theorem 1. *Let k be an odd positive integer. Then for all $n, m \geq 1$,*

$$\gcd(F(k^n) - 1, F(k^m) - 1) = F(k^{\gcd(n,m)} - 1).$$

For $k = 1$ every term is $F(1) - 1 = 0$ and the statement reads $\gcd(0, 0) = 0$; we therefore assume $k \geq 3$ throughout, in which case $k^n \geq 3$ and $a(k^n) \geq 1$, so all gcds below are gcds of positive integers.

2 Notation and four lemmas

Fix an odd $k \geq 3$. For $i \geq 1$ put

$$A_i = \frac{k^i - 1}{2}, \quad B_i = \frac{k^i + 1}{2},$$

both integers since k is odd. Then $B_i = A_i + 1$, so $\gcd(A_i, B_i) = 1$, and $A_i + B_i = k^i$.

Lemma 2 (uniform factorisation). *Let j be odd and let E and O be respectively the even and the odd member of $\{(j-1)/2, (j+1)/2\}$ (exactly one is even, since the two are consecutive). Then*

$$F(j) - 1 = F(E) L(O).$$

Proof. This is the classical factorisation of a difference of Fibonacci numbers: for $a \equiv b \pmod{4}$, $F(a) - F(b) = F(\frac{a-b}{2})L(\frac{a+b}{2})$, while for $a \equiv b+2 \pmod{4}$, $F(a) - F(b) = L(\frac{a-b}{2})F(\frac{a+b}{2})$; see for instance Vajda [3]. Take $b = 1$. If $j \equiv 1 \pmod{4}$ the first form gives $F(j) - 1 = F(\frac{j-1}{2})L(\frac{j+1}{2})$, and $\frac{j-1}{2}$ is even. If $j \equiv 3 \pmod{4}$ the second gives $F(j) - 1 = L(\frac{j-1}{2})F(\frac{j+1}{2})$, and $\frac{j+1}{2}$ is even. In both cases the Fibonacci factor carries the even index and the Lucas factor the odd one. $\square$

Lemma 3 (classical gcd laws). *Let $a, b \geq 1$ and $d = \gcd(a, b)$. Then*

$$(i) \quad \gcd(F(a), F(b)) = F(d);$$

$$(ii) \quad \text{if } a/d \text{ is even then } \gcd(F(a), L(b)) = L(d);$$

$$(iii) \quad \text{if } a/d \text{ and } b/d \text{ are both odd then } \gcd(L(a), L(b)) = L(d).$$

Proof. (i) is Vorob'ev's theorem. (ii) and (iii) are standard; see McDaniel [1] or Vajda [3]. In the complementary cases (ii) and (iii) give 1 or 2; we never need those branches, because every application below falls into the stated hypotheses. $\square$

Lemma 4 (gcds of $k^i \pm 1$). *Let $k \geq 3$ be odd, $n, m \geq 1$, $g = \gcd(n, m)$, $n' = n/g$, $m' = m/g$. Then*

$$\gcd(A_n, A_m) = A_g, \quad \gcd(B_n, B_m) = \begin{cases} B_g, & n', m' \text{ both odd,} \\ 1, & \text{otherwise,} \end{cases} \quad \gcd(A_n, B_m) = \begin{cases} B_g, & n' \text{ even,} \\ 1, & \text{otherwise.} \end{cases}$$

Proof. These are the standard cyclotomic gcd identities $\gcd(k^n - 1, k^m - 1) = k^g - 1$; $\gcd(k^n + 1, k^m + 1) = k^g + 1$ if n', m' are both odd and $= \gcd(2, k+1) = 2$ otherwise; and $\gcd(k^n - 1, k^m + 1) = k^g + 1$ if n' is even and $= 2$ otherwise, the value 2 arising because k is odd. Halving each identity gives the statement, since $\gcd(2X, 2Y) = 2 \gcd(X, Y)$. $\square$

Lemma 5 (parities). *Let $k \geq 3$ be odd, $g \mid n$, $n' = n/g$. Then, with $x = k^g$:*

$$(i) \quad A_n/A_g = 1 + x + \cdots + x^{n'-1} \text{ is odd if and only if } n' \text{ is odd};$$

- (ii) if n' is odd, $B_n/B_g = x^{n'-1} - x^{n'-2} + \dots + 1$ is odd;
- (iii) if n' is even, $A_n/B_g = x^{n'-1} - x^{n'-2} + \dots - 1$ is even;
- (iv) $k \equiv 1 \pmod{4}$ forces A_i even and B_i odd for every i ; $k \equiv 3 \pmod{4}$ forces A_i odd and B_i even for i odd, and A_i even and B_i odd for i even.

Proof. (i)–(iii): each displayed quotient is a sum of n' terms $\pm x^t$ with x odd, hence a sum of n' odd integers, whose parity is that of n' ; the identities themselves are the factorisations $X^{n'} - 1 = (X - 1)(1 + X + \dots)$ and, for the appropriate parity of n' , $X^{n'} + 1 = (X + 1)(X^{n'-1} - \dots + 1)$ and $X^{n'} - 1 = (X + 1)(X^{n'-1} - \dots - 1)$. (iv): $k \equiv 1 \pmod{4}$ gives $k^i \equiv 1$, so $4 \mid k^i - 1$ and A_i is even; $k \equiv 3 \pmod{4}$ gives $k^i \equiv 3$ for odd i and $\equiv 1$ for even i . $\square$

We also use twice the elementary fact that if $u_1 u_2 = u$ with $\gcd(u_1, u_2) = 1$, then $\gcd(u, v) = \gcd(u_1, v) \gcd(u_2, v)$.

3 Proof of Theorem 1

Fix odd $k \geq 3$ and $n, m \geq 1$; put $g = \gcd(n, m)$, $n' = n/g$, $m' = m/g$, so $\gcd(n', m') = 1$ and at most one of n', m' is even. If $n = m$ the statement is trivial, so assume $n \neq m$. By Lemma 2 applied to the odd numbers k^n, k^m, k^g , each of $a(k^n)$, $a(k^m)$, $a(k^g)$ is a product $F(\text{even index}) \cdot L(\text{odd index})$, the indices being A and B in one order or the other according to Lemma 5(iv).

Throughout, we use without further comment that $\gcd(A_i, B_i) = 1$ together with Lemma 3(ii) to get $\gcd(F(E_i), L(O_i)) = L(1) = 1$: the two factors of each $a(k^i)$ are coprime.

Case I: $k \equiv 1 \pmod{4}$

Here A_i is even and B_i odd for every i , so $a(k^i) = F(A_i)L(B_i)$.

I.1: n' and m' both odd. By Lemma 4, $\gcd(A_n, A_m) = A_g$ and $\gcd(B_n, B_m) = B_g$, while $\gcd(A_n, B_m) = \gcd(A_m, B_n) = 1$. Hence, by Lemma 3(i), $\gcd(F(A_n), F(A_m)) = F(A_g)$; by Lemma 3(iii) with Lemma 5(ii), $\gcd(L(B_n), L(B_m)) = L(B_g)$; and by Lemma 3(ii) (the Fibonacci index A_n being even and the gcd being 1) both cross terms $\gcd(F(A_n), L(B_m))$ and $\gcd(F(A_m), L(B_n))$ equal $L(1) = 1$.

Write $x = x_1 x_2$ and $y = y_1 y_2$ with $x_1 = F(A_n)$, $x_2 = L(B_n)$, $y_1 = F(A_m)$, $y_2 = L(B_m)$. We have $\gcd(x_1, x_2) = \gcd(y_1, y_2) = 1$ and $\gcd(x_1, y_2) = \gcd(x_2, y_1) = 1$. For a prime p dividing both x and y , p divides exactly one x_i and exactly one y_j , and the pairs (1, 2) and (2, 1) are excluded; hence $\gcd(x, y) = \gcd(x_1, y_1) \gcd(x_2, y_2) = F(A_g)L(B_g) = a(k^g)$.

I.2: exactly one of n', m' even. By symmetry assume n' even, m' odd. Then $\gcd(B_n, B_m) = 1$ and, both B 's being odd, Lemma 3(iii) gives $\gcd(L(B_n), L(B_m)) = L(1) = 1$. Also $\gcd(A_m, B_n) = 1$, so $\gcd(F(A_m), L(B_n)) = 1$. Therefore $L(B_n)$ is coprime to $F(A_m)L(B_m) = a(k^m)$, and

$$\gcd(a(k^n), a(k^m)) = \gcd(F(A_n), F(A_m)L(B_m)) = \gcd(F(A_n), F(A_m)) \cdot \gcd(F(A_n), L(B_m)),$$

the last step because $F(A_m)$ and $L(B_m)$ are coprime. The first factor is $F(A_g)$. For the second, $\gcd(A_n, B_m) = B_g$ by Lemma 4 (n' even), and A_n/B_g is even by Lemma 5(iii), so Lemma 3(ii) gives $L(B_g)$. Hence the gcd is $F(A_g)L(B_g) = a(k^g)$.

Case II: $k \equiv 3 \pmod{4}$

Now $a(k^i) = F(B_i)L(A_i)$ for odd i and $a(k^i) = F(A_i)L(B_i)$ for even i , by Lemma 5(iv).

II.a: n and m both odd. Then g is odd and n', m' are both odd. Here $a(k^n) = F(B_n)L(A_n)$, $a(k^m) = F(B_m)L(A_m)$ and $a(k^g) = F(B_g)L(A_g)$. Lemma 4 gives $\gcd(B_n, B_m) = B_g$, $\gcd(A_n, A_m) = A_g$ and $\gcd(B_n, A_m) = \gcd(B_m, A_n) = 1$ (both n', m' odd). So $\gcd(F(B_n), F(B_m)) = F(B_g)$; $\gcd(L(A_n), L(A_m)) = L(A_g)$ by Lemma 3(iii) with Lemma 5(i) (n', m' odd); and the two cross terms are $\gcd(F(B_n), L(A_m)) = L(1) = 1$ and likewise with n, m swapped, by Lemma 3(ii), the Fibonacci indices B_n, B_m being even. The same coprimality argument as in Case I.1 gives $\gcd(a(k^n), a(k^m)) = F(B_g)L(A_g) = a(k^g)$.

II.b: n and m both even. Then g is even, so k^n, k^m, k^g are all $\equiv 1 \pmod{4}$ and A_n, A_m, A_g are even while B_n, B_m, B_g are odd. Every ingredient of Case I — which used only these parities together with Lemmas 4 and 5 — applies verbatim, and gives $a(k^g)$.

II.c: exactly one of n, m odd. By symmetry assume n odd and m even. Since $g \mid n$, g is odd; since m is even and g odd, $m' = m/g$ is even, and therefore n' is odd (being coprime to m'). Thus

$$a(k^n) = F(B_n)L(A_n), \quad a(k^m) = F(A_m)L(B_m), \quad a(k^g) = F(B_g)L(A_g).$$

We compute the four pairings. Since n' is odd, $\gcd(A_n, B_m) = 1$ by Lemma 4, and A_n, B_m are both odd, so $\gcd(L(A_n), L(B_m)) = L(1) = 1$ by Lemma 3(iii). Since n' is odd and m' even, $\gcd(B_n, B_m) = 1$, and B_n is even, so $\gcd(F(B_n), L(B_m)) = L(1) = 1$ by Lemma 3(ii). Consequently $L(B_m)$ is coprime to $a(k^n)$, and

$$\gcd(a(k^n), a(k^m)) = \gcd(F(B_n)L(A_n), F(A_m)) = \gcd(F(B_n), F(A_m)) \cdot \gcd(L(A_n), F(A_m)),$$

using that $F(B_n)$ and $L(A_n)$ are coprime. For the first factor, m' is even so $\gcd(B_n, A_m) = B_g$ by Lemma 4, and Lemma 3(i) gives $F(B_g)$. For the second, $\gcd(A_m, A_n) = A_g$ and A_m/A_g is even by Lemma 5(i) (m' even), so Lemma 3(ii) gives $L(A_g)$. Hence the gcd is $F(B_g)L(A_g) = a(k^g)$. $\blacksquare$

4 Where the hypothesis is used

The oddness of k enters at three separate points, and the conclusion fails without it.

- A_i and B_i are integers only because k^i is odd. For even k the factorisation of Lemma 2 does not even apply, since $j = k^n$ is then even.
- The value 2 in the complementary branches of Lemma 4 is $\gcd(2, k+1)$, which is 2 exactly because k is odd; this is what makes the halved gcds equal to 1 and kills the cross terms.

- Lemma 5(i)–(iii) reads off parities from sums of n' odd powers of $x = k^g$; x odd is exactly the hypothesis.

For k even the statement is false at once: taking $k = 2$, $a(2) = F(2) - 1 = 0$, $a(4) = F(4) - 1 = 2$ and $a(8) = F(8) - 1 = 20$, so $\gcd(a(2^1), a(2^2)) = \gcd(0, 2) = 2$ while $a(2^{\gcd(1,2)}) = a(2) = 0$.

Remark 6. The theorem is not a statement about a itself. The sequence $a(j) = F(j) - 1$ is not a strong divisibility sequence: $\gcd(a(4), a(6)) = \gcd(2, 7) = 1$ while $a(\gcd(4, 6)) = a(2) = 0$. What the proof shows is that restricting the index to a fixed odd geometric progression $j = k^n$ converts a into one. The mechanism is visible in Lemma 2: along $j = k^n$ the two indices $(j \mp 1)/2$ become A_n, B_n , and these inherit the cyclotomic gcd structure of $k^n \mp 1$, which is what a strong divisibility sequence needs.

5 Computational verification

A script (Python 3, `sympy`) checks the following; all pass.

1. Lemma 2 for every odd $j < 400$.
2. The three branches of Lemma 3 used above, for all $1 \leq a, b \leq 59$.
3. Lemma 4 for all odd $3 \leq k < 30$ and $1 \leq n, m \leq 6$.
4. Every intermediate claim of the case analysis — the parities of A_i, B_i , the three gcds of Lemma 4, the quotient parities of Lemma 5, and the factorisation of each of $a(k^n)$, $a(k^m)$, $a(k^g)$ — on 102 triples (k, n, m) with k odd, $3 \leq k \leq 25$, $n, m \leq 6$ and $k^{\max(n,m)} \leq 1200$. All four regimes (I, II.a, II.b, II.c) are exercised, with 41, 17, 14 and 30 triples respectively.
5. The theorem itself on 151 triples (k, n, m) with k odd, $k < 40$, $n, m \leq 6$, $k^{\max(n,m)} \leq 900$, including $k = 1$.

Every numeral in this note is an output of that script.

6 Concluding remarks

The proof is elementary, but unlike most of what accumulates in OEIS comment fields it is not a one-liner: the content is the uniform factorisation of Lemma 2, after which the classical Fibonacci–Lucas gcd laws and the cyclotomic gcd identities do the work, and the case analysis is forced rather than chosen. We have found no earlier proof on the entry or in the literature on strong divisibility sequences, where the sequences studied are typically Lucas sequences themselves rather than shifts of them; but we would welcome notice of any earlier appearance.

References

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